LAPORAN PRAKTIKUM

POSTTEST (5)

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

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Kelas (B1‘25)

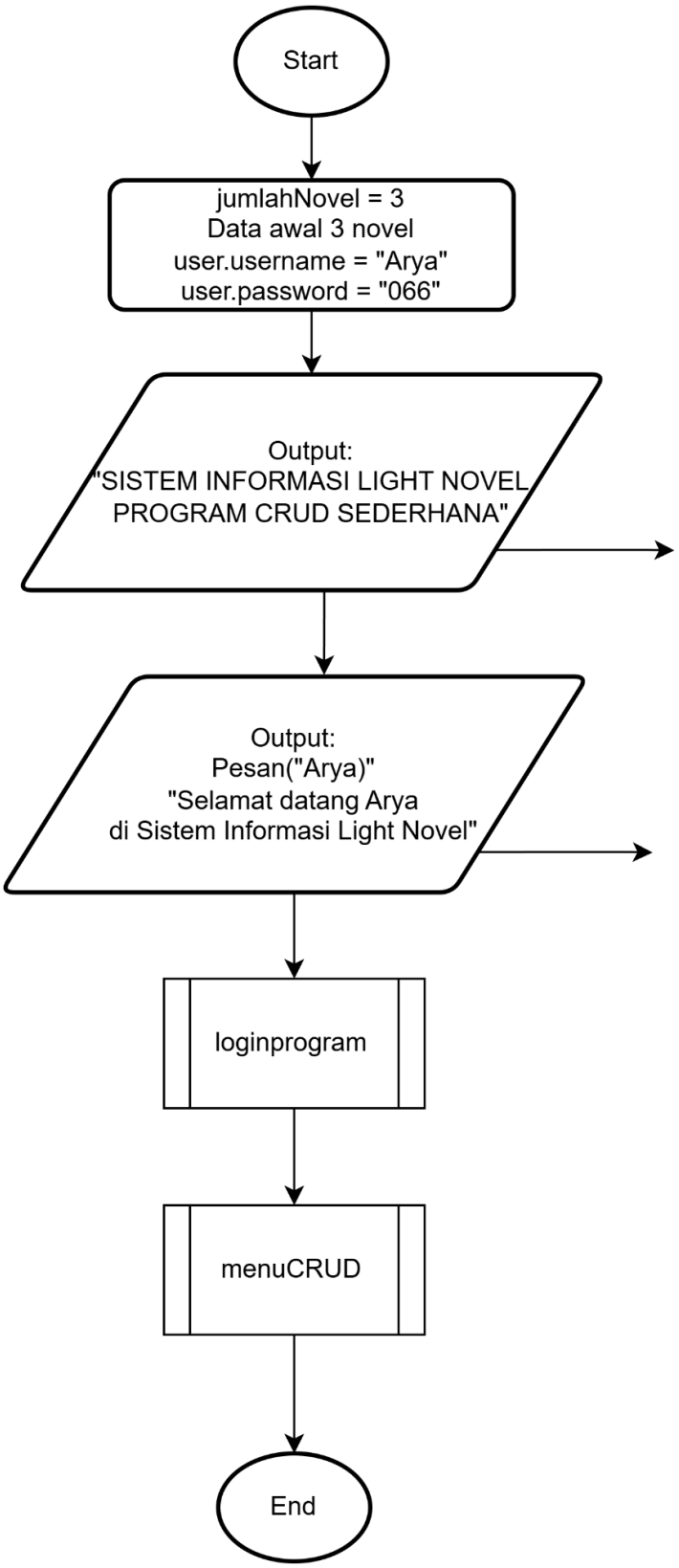
PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

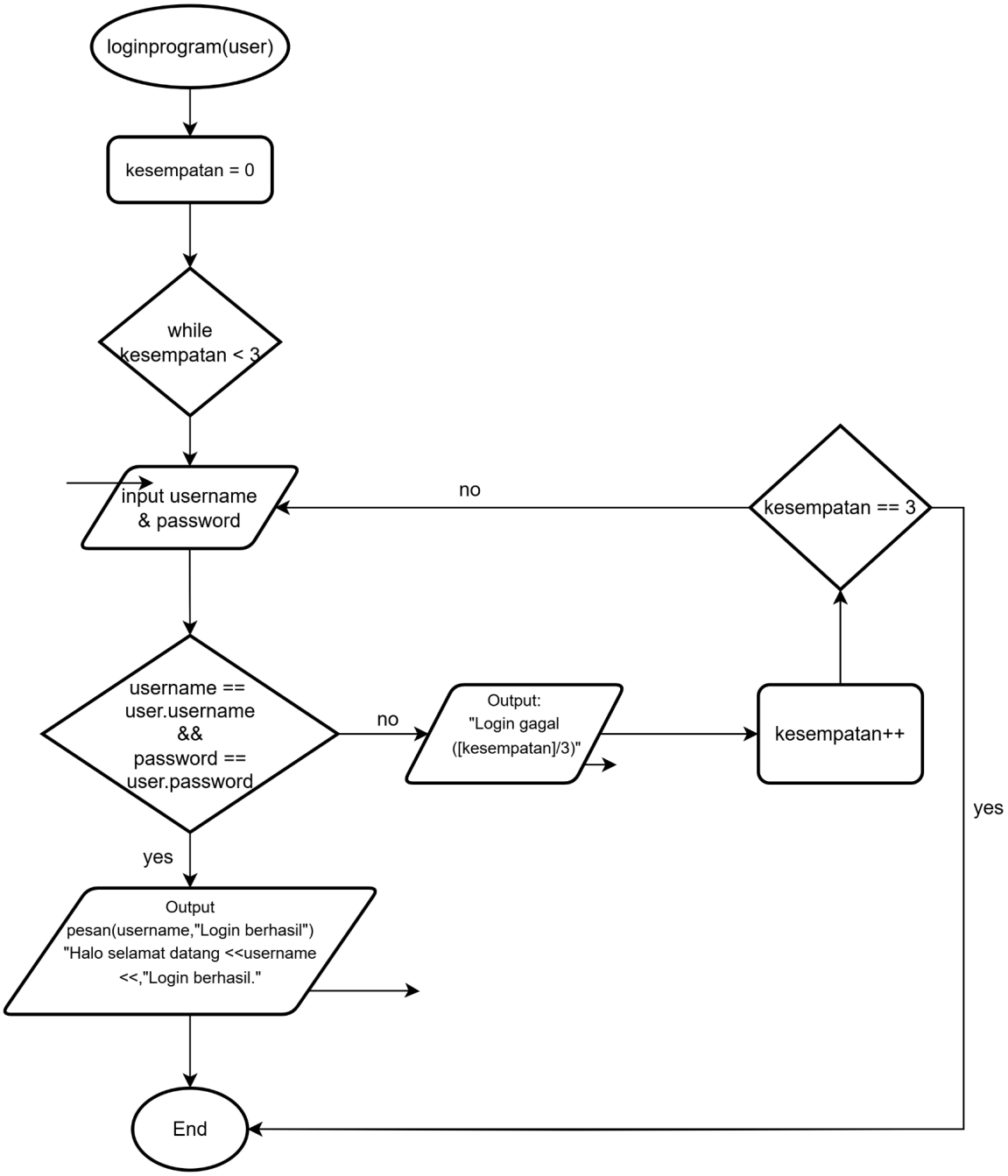
SAMARINDA

2025

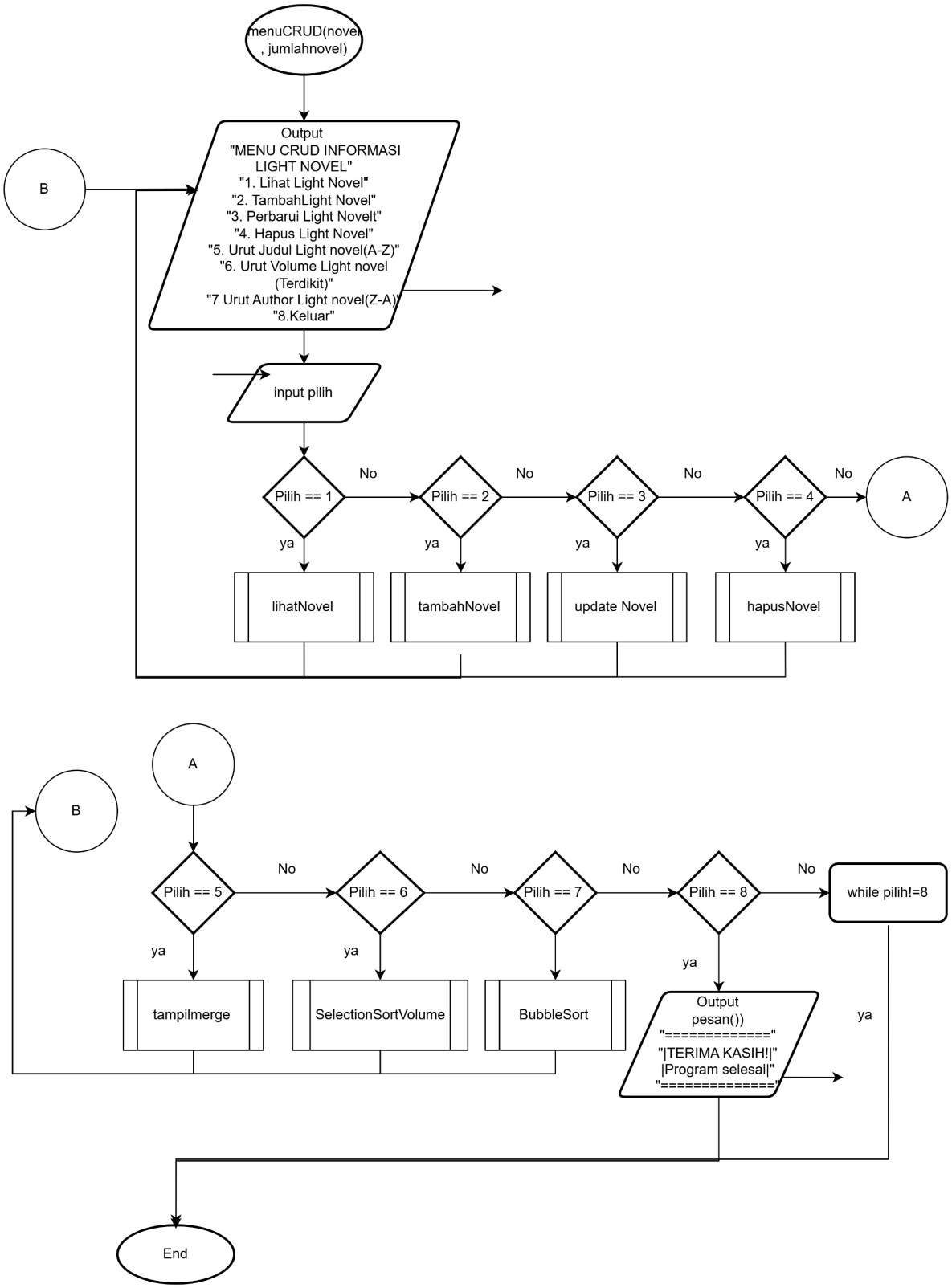
1. Flowchart



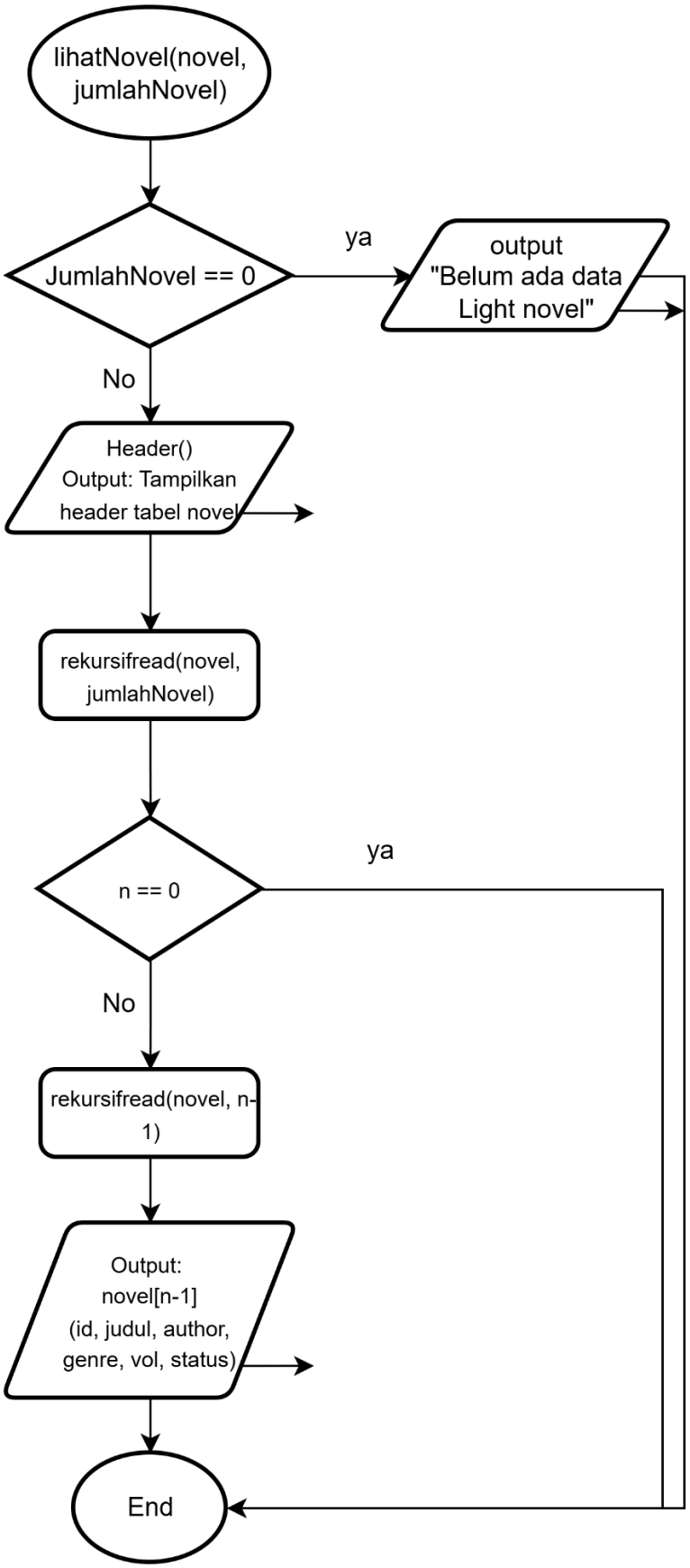
Gambar 1.1 Flowchart Main



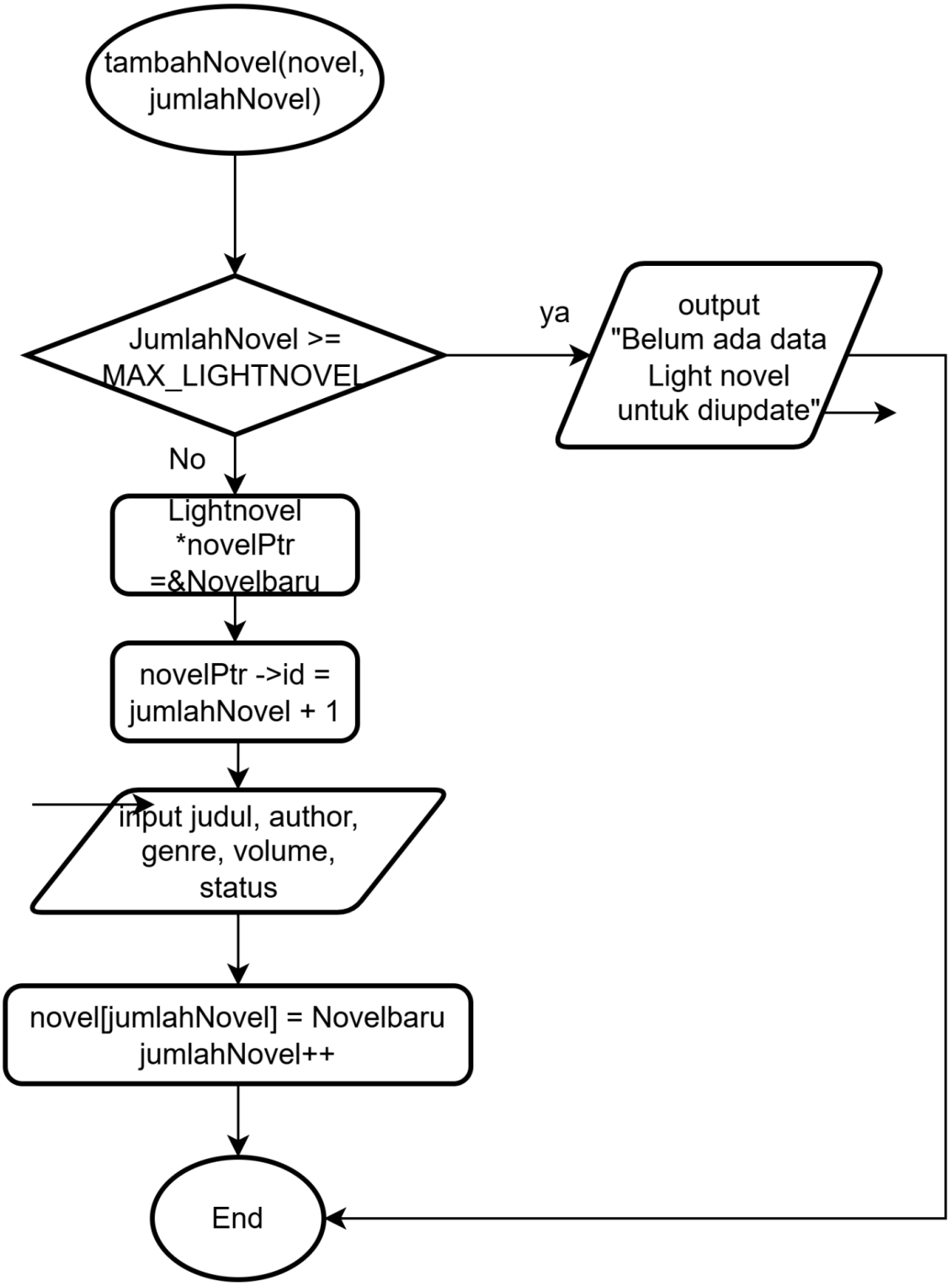
Gambar 1.2 Flowchart login program



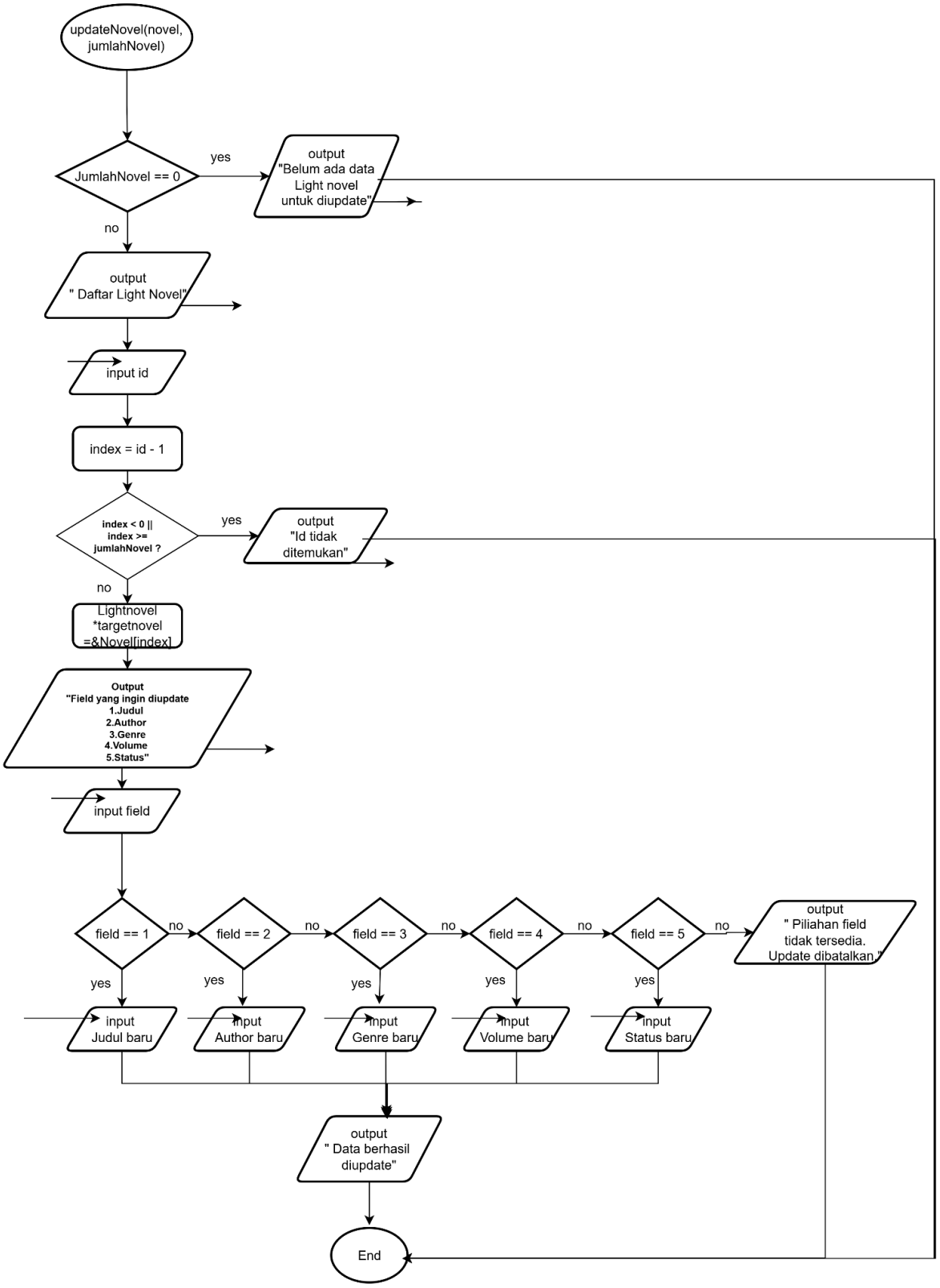
Gambar 1.3 Flowchart menu CRUD



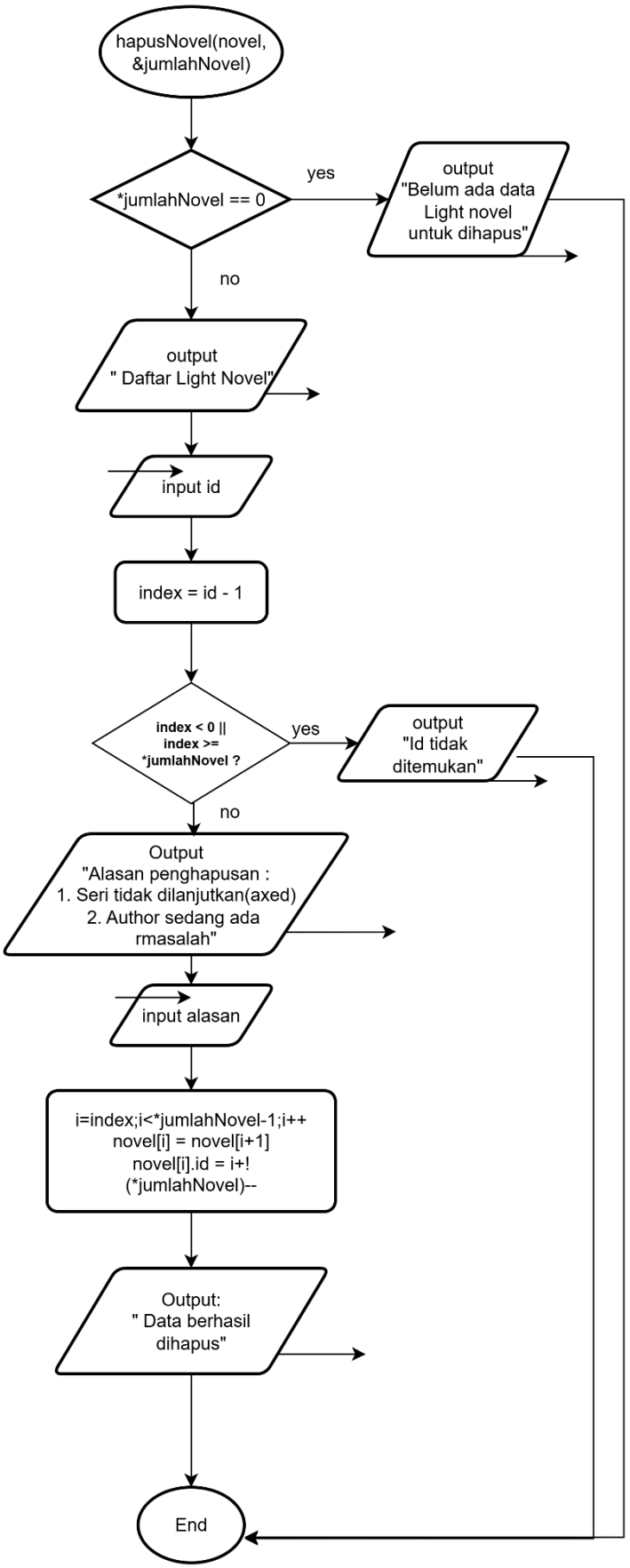
Gambar 1.4 Flowchart Read (lihatnovel)Informasi Light Novel



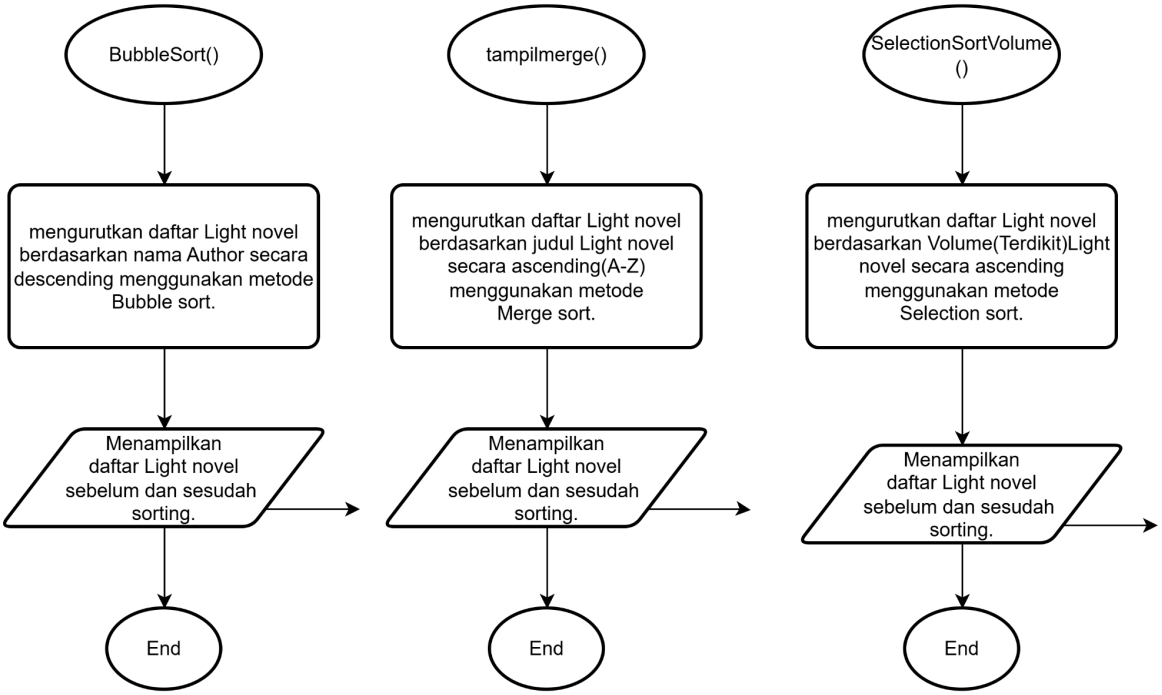
Gambar 1.5 Flowchart Create(tambahNovel)



Gambar 1.6 Flowchart Update(UpdateNovel)



Gambar 1.7 Flowchart Delete(hapusNovel)



Gambar 1.8 Flowchart Fitur sorting

Penjelasan Alur Flowchart :

1. **Inisialisasi Data Awal:** Sistem memulai dengan membuat satu akun admin (Arya, 066) dan mengisikan tiga data novel awal ke dalam array novel yang mampu menampung hingga 100 data (MAX_LIGHTNOVEL).

2. **Tampilan Menu Login:** Sistem menampilkan output awal dan meminta pengguna untuk mengisi username dan password melalui fungsi loginProgram.

3. **Verifikasi Login:** Fungsi loginProgram memeriksa input dari pengguna. Ada kesempatan hingga tiga kali percobaan. Apabila salah selama tiga kali, sistem akan secara otomatis berhenti (mengembalikan nilai false).

4. **Aktivasi Akses CRUD:** Setelah login berhasil (true), sistem akan memanggil fungsi menuCRUD untuk memberikan akses penuh ke menu pengelolaan data.

5. **Menu Utama (CRUD):** Sistem menyajikan menu yang berisi lima pilihan utama:
   1.Lihat Light Novel (Read)
   2.Tambah Light Novel (Create)
   3.Perbarui Light Novel (Update)
   4.Hapus Light Novel (Delete)
   5. Urut Judul Light novel A-Z(Merge Sort)
   6. Urut Volume Light novel dari tersedikit(Selection Sort)

7. Urut Author Light novel Z-A(Bubble Sort)
8.Keluar
ketika pengguna tidak memilih kelima menu diatas maka pilihan pengguna tidak valid dan akan kembali ke menu Crud lagi.

6. **Menu Read (Lihat Novel)**: Sistem menampilkan seluruh daftar novel yang tersimpan dengan menggunakan fungsi rekursif (rekursifread) dan menampilkannya dalam format tabel yang teratur dengan menggunakan setw (perataan teks).

7. **Menu Create (Tambah Novel):** Sistem meminta pengguna memasukkan detail novel baru (Judul, Author, Genre, Volume, Status) dan menyimpannya ke dalam array pada indeks terakhir yang tersedia, lalu memperbarui variabel jumlahNovel.

8. **Menu Update (Update Novel):** Sistem akan menampilkan daftar Light Novel dan meminta pengguna untuk memasukkan nomor ID novel, kemudian mencari indeks yang relevan. Jika ditemukan, sistem akan memberikan pengguna pilihan untuk memperbarui data yang ingin diubah/di update yaitu judul/author/genre/volume dan status. Jika memilih pilihan yang diinginkan, sistem akan meminta input baru terhadap pengguna untuk menggantikan data lama pada indeks tersebut.

9. **Menu Delete (Hapus Novel):** Sistem meminta nomor ID, lalu sistem akan menanyakan alasan penghapusan Light Novel. Kemudian sistem akan menghapus data tersebut dengan cara menggeser elemen array setelahnya ke posisi sebelumnya dan memperbarui urutan ID secara otomatis.
10. **Fitur Sorting** : Sistem Menyediakan 3 fitur sorting untuk memudahkan pengguna dalam mencari daftar light novel. Fitur sortingnya yaitu : mengurutkan Judul Light novel dari A-Z(Merge Sort), mengurutkan Volume dari yang terkecil/terdikit(Selection Sort), dan mengurutkan Author Light novel dari Z-A(Bubble Sort).

11. **Keluar :** Ketika pengguna memilih opsi ke 5 maka program keluar dari loop(selesai).

2. Deskripsi Singkat Program

Program ini merupakan program dengan tujuan untuk mengelola data informasi pada Light Novel dengan menggunakan sistem CRUD(Create,Read,Update,Delete) yang dibuat dengan menggunakan bahasa pemrograman C++ dan mengimplementasikan konsep pointer,fitur sorting(Bubble sort, merge sort,dan selection sort)pada program. Program ini dimulai dengan proses login yang membutuhkan pengguna untuk memasukkan username dan password yang benar. Pengguna diberi kesempatan maksimal tiga kali percobaan untuk bisa login. Jika sudah lebih dari tiga percobaan yang gagal, program akan otomatis berhenti.

Setelah proses login berhasil, pengguna bisa mengakses menu CRUD Informasi Light Novel yang terdiri dari:

1. Lihat Light Novel(Read).
2. Tambah Light Novel(Create).
3. Perbarui Light Novel(Update).
4. Hapus Light Novel(Delete).
5. Urut Judul Light novel A-Z(Merge Sort)
6. Urut Volume Light novel dari tersedikit(Selection Sort)
7. Urut Author Light novel Z-A(Bubble Sort)

pengguna dapat memilih menu untuk menambah, melihat, mengubah, atau menghapus **data** Light Novel, dan menu akan terus berulang sampai pengguna memilih keluar. Dengan adanya program ini, pengguna dapat mengelola data informasi Light Novel dengan lebih mudah, terstruktur, dan praktis.

3. Source Code

Source Code:

```
#include <iostream>

#include <iomanip>

#include <limits>


using namespace std;


#define MAX_LIGHTNOVEL 100


struct login {

    string username;

    string password;

};


struct detail {

    string genre;

    int volume;

    string status;

};
```

```
struct Lightnovel {
    int id;
    string judul;
    string author;
    detail detailNovel;
};

Lightnovel novel[MAX_LIGHTNOVEL] = {
    {1, "Sword Art Online", "Reki Kawahara", {"Fantasy", 27, "Ongoing"}},
    {2, "Overlord", "Kugane Maruyama", {"Isekai", 17, "Ongoing"}},
        {3, "Classroom of the Elite", "Syougo Kinugasa", {"School", 31,
"Ongoing"}}
};

int jumlahNovel = 3;

void clearBuffer() {
    cin.clear();
    cin.ignore(numeric_limits<streamsize>::max(), '\n');
}

//Overloading
string Pesan(string nama) {
```

```
      string tx = "Selamat datang " + nama + " di Sistem Informasi Light
Novel.\n";

    cout << tx;

    return tx;

}


string Pesan(string nama, string status) {

    string tx = "Halo " + nama + ", " + status;

    cout<<tx;

    return tx;

}


void Header(string judul) {

    cout << "\n" << judul << "\n";

                                                                cout              <<
"================================================================================
===================\n";

    cout << "|" << setw(4) << "ID" << " | ";

    cout << setw(25) << "Judul" << " | ";

    cout << setw(20) << "Author" << " | ";

    cout << setw(12) << "Genre" << " | ";

    cout << setw(6) << "Vol" << " | ";

    cout << setw(12) << "Status" << " |\n";

                                                                cout              <<
"================================================================================
===================\n";
```

```
}
void tukarnovel(Lightnovel* a, Lightnovel* b) {
    Lightnovel temp = *a;
    *a = *b;
    *b = temp;
}

//rekursif
int rekursifread(Lightnovel novel[], int n) {
    if (n == 0) return 0;
    rekursifread(novel, n - 1);
    cout << "|" << setw(4) << novel[n - 1].id << " | ";
    cout << setw(25) << novel[n - 1].judul << " | ";
    cout << setw(20) << novel[n - 1].author << " | ";
    cout << setw(12) << novel[n - 1].detailNovel.genre << " | ";
    cout << setw(6) << novel[n - 1].detailNovel.volume << " | ";
    cout << setw(12) << novel[n - 1].detailNovel.status << " |\n";
    return n;
}

void lihatNovel(Lightnovel *novel, int jumlahNovel, bool clear = true) {
    if(clear)system("cls");
    if (jumlahNovel == 0) {
        cout << "Belum ada data novel.\n";
```

```
        return;
    }
    Header(" Daftar Light Novel :\n");
    rekursifread(novel, jumlahNovel);
                                                        cout              <<
"================================================================================
===================\n\n";
}


int tambahNovel(Lightnovel novel[], int &jumlahNovel) {
    system("cls");
    if (jumlahNovel >= MAX_LIGHTNOVEL) {
        cout << "Data novel penuh\n";
        return 0;
    }


    Lightnovel Novelbaru;
    Lightnovel *novelPtr = &Novelbaru;
    novelPtr->id = jumlahNovel + 1;
    cout<< "tekan Enter untuk lanjut...";
    clearBuffer();
    cout << "Masukkan Judul Light Novel : ";
    getline(cin, novelPtr->judul);
    cout << "Masukkan Author            : ";
    getline(cin, novelPtr->author);
```

```
    cout << "Masukkan Genre             : ";
    getline(cin, novelPtr->detailNovel.genre);
    cout << "Masukkan Volume            : ";
    cin >> novelPtr->detailNovel.volume;
    clearBuffer();
    cout << "Masukkan Status (Ongoing/Completed) : ";
    getline(cin, novelPtr->detailNovel.status);
    novel[jumlahNovel] = Novelbaru;
    jumlahNovel++;
    return 1;
}

void updateNovel(Lightnovel *novel, int jumlahNovel) {
    system("cls");
    if(jumlahNovel == 0){
        cout<<"\nBelum ada data Light Novel untuk diupdate.\n"<<endl;
    }
    else{
        int id;
        Header(" Daftar Light Novel :\n");
        for(int i=0;i<jumlahNovel;i++){
            cout<<"|";
            cout<<setw(4)<<novel[i].id<<" | ";
            cout<<setw(25)<<novel[i].judul<<" | ";
```

```
            cout<<setw(20)<<novel[i].author<<" | ";

            cout<<setw(12)<<novel[i].detailNovel.genre<<" | ";

            cout<<setw(6)<<novel[i].detailNovel.volume<<" | ";

            cout<<setw(12)<<novel[i].detailNovel.status<<" |\n";

        }


cout<<"==========================================================================
==========================\n";

        cout<<"\nMasukkan ID Light Novel yang ingin diupdate : ";

        cin>>id;

        int index = id - 1;

        if(index < 0 || index >= jumlahNovel){

            cout<<"ID tidak ditemukan.\n";

        }

        else{

            Lightnovel *targetNovel = &novel[index];

            int field;

            cout<<"Field yang ingin diupdate\n";

            cout<<"1. Judul\n";

            cout<<"2. Author\n";

            cout<<"3. Genre\n";

            cout<<"4. Volume\n";

            cout<<"5. Status\n";

            cout<<"Pilih : ";

            cin>>field;
```

```
        cin.ignore(1000,'\n');
        if(field == 1){
            cout<<"Masukkan Judul Baru : ";
            getline(cin, targetNovel->judul);
            cout<<"Data berhasil diupdate.\n"<<endl;
        }
        else if(field == 2){
            cout<<"Masukkan Author Baru : ";
            getline(cin, targetNovel->author);
            cout<<"Data berhasil diupdate.\n"<<endl;
        }
        else if(field == 3){
            cout<<"Masukkan Genre Baru : ";
            getline(cin, targetNovel->detailNovel.genre);
            cout<<"Data berhasil diupdate.\n"<<endl;
        }
        else if(field == 4){
            cout<<"Masukkan Volume Baru : ";
            cin>>targetNovel->detailNovel.volume;
            cout<<"Data berhasil diupdate.\n"<<endl;
        }
        else if(field == 5){
            cout<<"Masukkan Status Baru : ";
            getline(cin, targetNovel->detailNovel.status);
```

```
                cout<<"Data berhasil diupdate.\n"<<endl;
            }
            else{
                cout<<"Pilihan field tidak tersedia. Update dibatalkan.\n";
            }
        }
    }
}


int hapusNovel(Lightnovel novel[], int *jumlahNovel) {
    system("cls");
    if(*jumlahNovel == 0){
        cout<<"\nBelum ada data Light Novel untuk dihapus.\n"<<endl;
        return 0;
    }
    else{
        Header(" Daftar Light Novel :\n");
        for(int i=0;i<*jumlahNovel;i++){
            cout<<"|";
            cout<<setw(4)<<novel[i].id<<" | ";
            cout<<setw(25)<<novel[i].judul<<" | ";
            cout<<setw(20)<<novel[i].author<<" | ";
            cout<<setw(12)<<novel[i].detailNovel.genre<<" | ";
            cout<<setw(6)<<novel[i].detailNovel.volume<<" | ";
```

```
            cout<<setw(12)<<novel[i].detailNovel.status<<" |\n";

        }

cout<<"==========================================================================
==========================\n";


        int id;

        cout<<"Masukkan ID yang ingin dihapus : ";

        cin>>id;

        int index = id - 1;

        if(index < 0 || index >= *jumlahNovel){

            cout<<"ID tidak ditemukan.\n"<<endl;

            return 0;

        }

        else{

            int alasan;

            cout<<"\nAlasan penghapusan:\n";

            cout<<"1. Seri tidak dilanjutkan (Axed)\n";

            cout<<"2. Author sedang ada masalah\n";

            cout<<"Pilih alasan : ";

            cin>>alasan;


            for(int i=index;i<*jumlahNovel-1;i++){

                novel[i] = novel[i+1];

                novel[i].id = i+1;
```

```
            }
            (*jumlahNovel)--;
            cout<<"\nData berhasil dihapus.\n\n"<<endl;
            return 1;
        }
    }
}
void merge(Lightnovel* arr, int left, int mid, int right) {
    int n1 = mid - left + 1;
    int n2 = right - mid;

    Lightnovel* L = new Lightnovel[n1];
    Lightnovel* R = new Lightnovel[n2];

    for (int i = 0; i < n1; i++)
        *(L + i) = *(arr + left + i);
    for (int j = 0; j < n2; j++)
        *(R + j) = *(arr + mid + 1 + j);
    int i = 0, j = 0, k = left;
    while (i < n1 && j < n2) {
        if ((L + i)->judul <= (R + j)->judul) {
            *(arr + k) = *(L + i);
            i++;
        } else {
```

```
            *(arr + k) = *(R + j);
            j++;
        }
        k++;
    }
    while (i < n1) {
        *(arr + k) = *(L + i);
        i++; k++;
    }
    while (j < n2) {
        *(arr + k) = *(R + j);
        j++; k++;
    }
    delete[] L;
    delete[] R;
}

void mergeSort(Lightnovel* arr, int left, int right) {
    if (left < right) {
        int mid = left + (right - left) / 2;
        mergeSort(arr, left, mid);
        mergeSort(arr, mid + 1, right);
        merge(arr, left, mid, right);
    }
```

```
}

void tampilmerge(Lightnovel* arr, int left, int right) {

     system("cls");

     if (jumlahNovel == 0) {

                    cout << "Belum ada data Light novel." << endl;

                    return;

                }

                   cout << "Data sebelum diurutkan berdasarkan judul(A-Z)" <<
endl;

                lihatNovel(novel, jumlahNovel, false);

                mergeSort(novel, 0, jumlahNovel - 1);

                  cout << "Data berhasil diurutkan berdasarkan judul(A-Z)!" <<
endl;

                lihatNovel(novel, jumlahNovel,false);

}


void selectionSortVolume(Lightnovel* arr, int n) {

    system("cls");

    if (n == 0) {

        cout << "Belum ada data Light novel." << endl;

        return;

    }

    cout << "Data sebelum diurutkan berdasarkan Volume(Terdikit)" << endl;

    lihatNovel(novel, jumlahNovel, false);
```

```
    for (int i = 0; i < n - 1; i++) {
        int minIdx = i;
        for (int j = i + 1; j < n; j++) {
                                if  ((arr  +  j)->detailNovel.volume  <  (arr  +
minIdx)->detailNovel.volume) {
                minIdx = j;
            }
        }
        if (minIdx != i) {
            tukarnovel(arr + i, arr + minIdx);
        }
    }
    cout << "Data berhasil diurutkan berdasarkan Volume (Terdikit)!" << endl;
    lihatNovel(arr, n, false);
}


void bubbleSort(Lightnovel *arr, int n) {
    system("cls");
    if (n ==0){
        cout << "Belum ada data Light novel." << endl;
        return;
    }
    cout << "Data sebelum diurutkan berdasarkan Author (Z-A)" << endl;
    lihatNovel(novel, jumlahNovel, false);
```

```
    bool swapped;
    for (int i = 0; i < n - 1; i++) {
        swapped = false;
        for (int j = 0; j < n - i - 1; j++) {
            if ((arr + j)->author < (arr + j + 1)->author) {
                tukarnovel(arr + j, arr + j + 1);
                swapped = true;
            }
        }
        if (swapped == false)
            break;
    }
    cout << "Data berhasil diurutkan berdasarkan Author (Z-A)!" << endl;
    lihatNovel(arr, n, false);
}
bool loginProgram(login user) {
    string username, password;
    int kesempatan = 0;
    while (kesempatan < 3) {
        cout << "Username : "; cin >> username;
        cout << "Password : "; cin >> password;
        if (username == user.username && password == user.password) {
            Pesan(username, "Login berhasil ");
            system("pause");
```

```
            return true;
        }
        kesempatan++;
        cout << "Login gagal (" << kesempatan << "/3)\n";
    }
    return false;
}

void menuCRUD(Lightnovel novel[], int &jumlahNovel) {
    int pilih;
    do {
        system("cls");
        cout << "========================================\n";
        cout << "|    MENU CRUD INFORMASI LIGHT NOVEL   |\n";
        cout << "========================================\n";
        cout << "|1. Lihat Light Novel                  |\n";
        cout << "|2. Tambah Light Novel                 |\n";
        cout << "|3. Perbarui Light Novel               |\n";
        cout << "|4. Hapus Light Novel                  |\n";
        cout << "|5. Urut Judul Light Novel(A-Z)        |\n";
        cout << "|6. Urut Volume Light Novel(Terdikit)  |\n";
        cout << "|7. Urut Author Light novel(Z-A)       |\n";
        cout << "|8. Keluar                             |\n";
        cout << "========================================\n";
```

```
        cout << "Pilih menu : ";


        cin >> pilih;
        clearBuffer();


        switch (pilih) {
            case 1:
                lihatNovel(novel, jumlahNovel);
                system("pause");
                break;
            case 2:
                if (tambahNovel(novel, jumlahNovel)) {
                    Pesan("Admin", "Data berhasil ditambahkan ");
                }
                system("pause");
                break;
            case 3:
                updateNovel(novel, jumlahNovel);
                system("pause");
                break;
            case 4:
                if (hapusNovel(novel, &jumlahNovel)) {
                    Pesan("Admin", "Data berhasil dihapus ");
                }
```

```
            system("pause");
            break;
        case 5:
            tampilmerge(novel, 0, jumlahNovel - 1);
            system("pause");
            break;
        case 6:
            selectionSortVolume(novel, jumlahNovel);
            system("pause");
            break;
        case 7:
            bubbleSort(novel, jumlahNovel);
            system("pause");
            break;
        case 8:
            system("cls");
            cout << "================================\n";
            cout << "|         TERIMA KASIH!        |\n";
            cout << "|        Program selesai       |\n";
            cout << "================================\n" << endl;
            break;
        default:
            cout << "Menu tidak tersedia\n";
            system("pause");
```

```
        }
    } while (pilih != 8);
}


int main(){
    login user = {"Arya","066"};
    cout<<"====================================\n";
    cout<<"|   SISTEM INFORMASI LIGHT NOVEL   |\n";
    cout<<"|      PROGRAM CRUD SEDERHANA      |\n";
    cout<<"====================================\n"<<endl;
    Pesan("Arya");
    if(loginProgram(user)){
        menuCRUD(novel,jumlahNovel);
    }
    return 0;
}
```

4. Hasil Output

```
=====================================
|   SISTEM INFORMASI LIGHT NOVEL    |
|      PROGRAM CRUD SEDERHANA       |
=====================================

Selamat datang Arya di Sistem Informasi Light Novel.
Username : Arya
Password : 066
Halo Arya, Login berhasil Press any key to continue . . .
```

Gambar 4.1 Tampilan login berhasil

```
=====================================
|   SISTEM INFORMASI LIGHT NOVEL    |
|      PROGRAM CRUD SEDERHANA       |
=====================================

Selamat datang Arya di Sistem Informasi Light Novel.
Username : df
Password : fss
Login gagal (1/3)
Username : fsf
Password : sfsf
Login gagal (2/3)
Username : sfsf
Password : sfsf
Login gagal (3/3)
```

Gambar 4.2 Tampilan jika login gagal dan melebihi 3 kali percobaan atau kesempatan.

```
=========================================
|    MENU CRUD INFORMASI LIGHT NOVEL    |
=========================================
|1. Lihat Light Novel                   |
|2. Tambah Light Novel                  |
|3. Perbarui Light Novel                |
|4. Hapus Light Novel                   |
|5. Urut Judul Light Novel(A-Z)         |
|6. Urut Volume Light Novel(Terdikit)   |
|7. Urut Author Light novel(Z-A)        |
|8. Keluar                              |
=========================================
Pilih menu :
```

Gambar 4.3 Tampilan Menu Crud Informasi Light novel

```
 Daftar Light Novel :

=================================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
=================================================================================================
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     17 |      Ongoing |
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     31 |      Ongoing |
=================================================================================================

Press any key to continue . . .
```

Gambar 4.4 Output pada Read Informasi Light Novel

```
tekan Enter untuk lanjut
Masukkan Judul Light Novel : Ryuu Kusari no Ori
Masukkan Author            : Cadet
Masukkan Genre             : Fantasy
Masukkan Volume            : 4
Masukkan Status (Ongoing/Completed) : Completed
Halo Admin, Data berhasil ditambahkan.
Press any key to continue . . .
```

Gambar 4.5 Output pada Create Informasi Light Novel

```
Daftar Light Novel:
=================================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
=================================================================================================
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     16 |      Ongoing |
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     14 |      Ongoing |
|   4 |        Ryuu Kusari no Ori |                Cadet |      Fantasy |      4 |    Completed |
=================================================================================================

Masukkan ID Light Novel yang ingin diupdate : 5
ID tidak ditemukan.
```

Gambar 4.6.1 Output pada Update Informasi Light Novel jika id Light Novel tidak ditemukan

```
Daftar Light Novel:
==========================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
==========================================================================================
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     16 |      Ongoing |
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     14 |      Ongoing |
|   4 |        Ryuu Kusari no Ori |                Cadet |      Fantasy |      4 |    Completed |
==========================================================================================

Masukkan ID Light Novel yang ingin diupdate : 4

Field yang ingin diupdate
1. Judul
2. Author
3. Genre
4. Volume
5. Status
Pilih : 3
Masukkan Genre Baru : Action
Data berhasil diupdate.
```

Gambar 4.6.2 Output pada Update Informasi Light Novel

```
Daftar Light Novel:
==========================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
==========================================================================================
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     17 |      Ongoing |
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     31 |      Ongoing |
|   4 |        Ryuu Kusari no Ori |                Cadet |       Action |      4 |    Completed |
==========================================================================================
Masukkan ID yang ingin dihapus : 5
ID tidak ditemukan.

Press any key to continue . . .
```

Gambar 4.7.1 Output pada Delete Informasi Light Novel jika id Light Novel tidak ditemukan

```
Daftar Light Novel:
==================================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
==================================================================================================
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     17 |      Ongoing |
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     31 |      Ongoing |
|   4 |        Ryuu Kusari no Ori |                Cadet |       Action |      4 |    Completed |
==================================================================================================
Masukkan ID yang ingin dihapus : 4

Alasan penghapusan:
1. Seri tidak dilanjutkan (Axed)
2. Author sedang ada masalah
Pilih alasan : 1

Data berhasil dihapus.


Halo Admin, Data berhasil dihapus.
Press any key to continue . . .
```

Gambar 4.7.2 Output pada Delete Informasi Light Novel

```
Data sebelum diurutkan berdasarkan judul(A-Z)

 Daftar Light Novel :

==================================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
==================================================================================================
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     17 |      Ongoing |
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     31 |      Ongoing |
==================================================================================================

Data berhasil diurutkan berdasarkan judul(A-Z)!

 Daftar Light Novel :

==================================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
==================================================================================================
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     31 |      Ongoing |
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     17 |      Ongoing |
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
==================================================================================================

Press any key to continue . . .
```

Gambar 4.8 Output pada Sorting Judul Light Novel A-Z(Merge Sort)

```
Data sebelum diurutkan berdasarkan Volume(Terdikit)

 Daftar Light Novel :

==========================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
==========================================================================================
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     17 |      Ongoing |
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     31 |      Ongoing |
==========================================================================================

Data berhasil diurutkan berdasarkan Volume (Terdikit)!

 Daftar Light Novel :

==========================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
==========================================================================================
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     17 |      Ongoing |
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     31 |      Ongoing |
==========================================================================================

Press any key to continue . . .
```

Gambar 4.9 Output pada Sorting Volume Light Novel Tersedikit(Selection Sort)

```
Data sebelum diurutkan berdasarkan Author (Z-A)

 Daftar Light Novel :

==========================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
==========================================================================================
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     17 |      Ongoing |
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     31 |      Ongoing |
==========================================================================================

Data berhasil diurutkan berdasarkan Author (Z-A)!

 Daftar Light Novel :

==========================================================================================
|  ID |                    Judul |              Author |        Genre |    Vol |       Status |
==========================================================================================
|   3 |    Classroom of the Elite |      Syougo Kinugasa |       School |     31 |      Ongoing |
|   1 |          Sword Art Online |        Reki Kawahara |      Fantasy |     27 |      Ongoing |
|   2 |                  Overlord |      Kugane Maruyama |       Isekai |     17 |      Ongoing |
==========================================================================================

Press any key to continue . . .
```

Gambar 4.10 Output pada Sorting Author Light Novel Z-A(Bubble Sort)

```
=========================================
|    MENU CRUD INFORMASI LIGHT NOVEL    |
=========================================
|1. Lihat Light Novel                   |
|2. Tambah Light Novel                  |
|3. Perbarui Light Novel                |
|4. Hapus Light Novel                   |
|5. Urut Judul Light Novel(A-Z)         |
|6. Urut Volume Light Novel(Terdikit)   |
|7. Urut Author Light novel(Z-A)        |
|8. Keluar                              |
=========================================
Pilih menu : 9
Menu tidak tersedia
Press any key to continue . . .
```

Gambar 4.11 Tampilan jika menu Crud tidak tersedia

```
================================
|         TERIMA KASIH!        |
|        Program selesai       |
================================
```

Gambar 4.12 Tampilan jika keluar dari program

5. Langkah-langkah GIT

5.1 GIT add:

```
PS C:\Users\user\Documents\praktikum-apl> git add .
```

Gambar 5.1 git add

git add untuk menginisiasi atau menambahkan file yang sudah diubah atau baru di repository.

5.2 GIT Commit :

```
PS C:\Users\user\Documents\praktikum-apl> git commit -m "testing"
[main 1832234] testing
 3 files changed, 26 insertions(+), 13 deletions(-)
```

Gambar 5.2 git commit

git commit untuk menyimpan perubahan repository dengan pesan commit.

5.3 GIT Push :

```
PS C:\Users\user\Documents\praktikum-apl> git push -u origin main
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 4 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (8/8), 237.38 KiB | 1.14 MiB/s, done.
Total 8 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 5 local objects.
To https://github.com/AryasatyaITX/praktikum-apl.git
```

Gambar 5.3 git push

git push berfungsi untuk mengirim commit dari repository lokal ke repository remote (GitHub) sekaligus menetapkan branch default.